Mid-Term Assignment (Fall 2026)

Overview

This mid-term assignment consists of two main parts designed to assess your understanding of XML processing and relational database interactions. You may use **any programming language of your choice** (e.g., Java, Python, C#, Node.js, C++, etc.) to complete the tasks.

Part1: XML Parsing and Database Persistence

Write a program to read and parse the XML file shown in **Figure 1** (score_data.xml). After parsing, insert the student records into a MySQL database table named score_data under the schema xmldb (as shown in **Figure 2**).

```
<?xml version="1.0"?>
<score_data>
    <student student_id="A001">
        <xml_class>60</xml_class>
        <data_structure>70</data_structure>
        <algorithm>85</algorithm>
        <network>90</network>
    </student>
    <student student_id="A002">
        <xml_class>66</xml_class>
        <data_structure>78</data_structure>
        <algorithm>62</algorithm>
        <network>88</network>
    </student>
    <student student_id="A003">
        <xml_class>89</xml_class>
        <data_structure>77</data_structure>
        <algorithm>80</algorithm>
        <network>50</network>
    </student>
</score_data>
```

Figure 1: score_data.xml

Database Schema Details:

- Database / Schema: xmldb
- Table Name: score_data
- Columns & Types:
    - student_id: VARCHAR(5) (Primary Key, Not Null)
    - xml_score: INT
    - data_structure_score: INT
    - algorithm_score: INT
    - network_score: INT

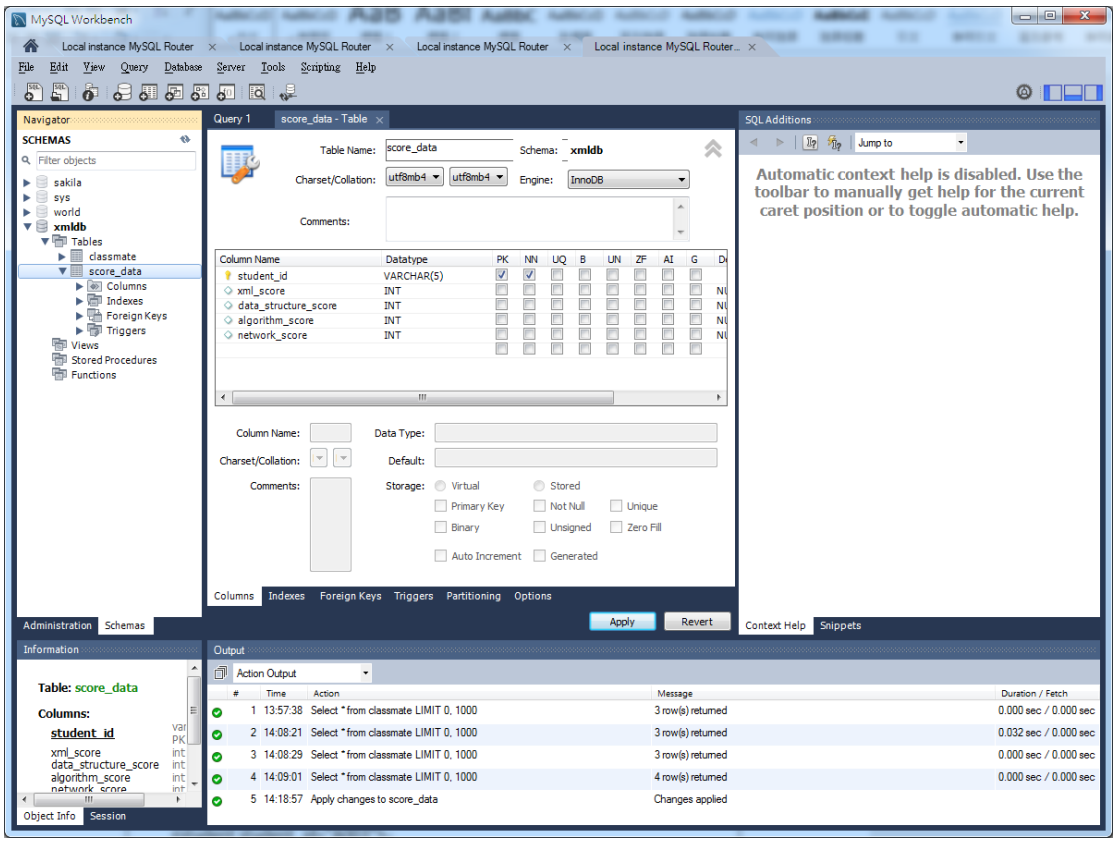


Figure 2: MySQL Workbench table structure and configurations

Part 2: Data Retrieval, GPA Calculation, and XML Generation

Write a program to perform the following steps:

1. Query and retrieve all student records from the score_data database table.
2. Calculate the corresponding Grade Point (GP) for each course score based on the **Grading Scale (Table 1)**.
3. Construct a new XML document in memory where each course element includes a gpa attribute indicating the converted GP value.
4. Export and save the updated XML tree into a new file named output.xml (as

illustrated in **Figure 3**).

```
<?xml version="1.0"?>
<score_data>
    <student student_id="A001">
        <xml_class gpa="2">60</xml_class>
        <data_structure gpa="2.7">70</data_structure>
        <algorithm gpa="4">85</algorithm>
        <network gpa="4.5">90</network>
    </student>
    <student student_id="A002">
        <xml_class gpa="2.3">66</xml_class>
        <data_structure gpa="3.3">78</data_structure>
        <algorithm gpa="2">62</algorithm>
        <network gpa="4">88</network>
    </student>
    <student student_id="A003">
        <xml_class gpa="4">89</xml_class>
        <data_structure gpa="3.3">77</data_structure>
        <algorithm gpa="3.7">80</algorithm>
        <network gpa="1">50</network>
    </student>
</score_data>
```

Figure 3: output.xml

Table 1: Score to Grade Point (GP) Conversion Scale

積分(GP值)	百分計分法	等第記分法	積分(GP值)	百分計分法	等第記分法
4.5	90-100	A+	2.5	67-69	C+
4.0	85-89	A	2.3	63-66	C
3.7	80-84	A-	2.0	60-62	C-
3.3	77-79	B+	1.0	50-59	D
3.0	73-76	B	0.0	<50	E
2.7	70-72	B-			

Submission Requirements

1. **Code & Files Submission (Zip Archive):**
    - Package all your deliverables into a single .zip file named [YourStudentID]_Midterm.zip and upload it to the course platform.
    - Your archive must include:
        - Complete source code files for Part 1 and Part 2 (in whatever programming language you used).
        - The input XML file (score_data.xml) and the generated output XML file (output.xml).
        - Any necessary configuration or dependency files (e.g., requirements.txt, pom.xml, package.json, or environment setup notes).
        - A README.txt file containing the shareable link to your video presentation along with brief instructions on how to run your code.
2. **Video Presentation (Demo Video):**
    - Record a concise video (around 3–5 minutes) explaining and demonstrating your work.
    - Required Video Content:
        - A brief walk-through of your source code logic for Part 1 and Part 2.
        - Demonstration of running Part 1, verifying that data from score_data.xml is successfully written to the MySQL database.
        - Demonstration of running Part 2, verifying that GP values are correctly computed and output.xml is generated.
    - Submission Method:
        - Upload your video to YouTube (set visibility to "Unlisted") or Google Drive (set permissions to "Anyone with the link can view").
        - Paste the accessible video link into the README.txt file inside your uploaded zip package.